communicate with us."[†] Within a few days, however, she discovered other repeating signals coming from other places in our Milky Way galaxy. Bell and her associates realized they'd discovered a new class of cosmic object—a star made entirely of neutrons that pulses with radio waves for every rotation it executes. Hewish and Bell sensibly named them "pulsars."

Turns out, intercepting radio waves isn't the only way to be snoopy. There's also cosmochemistry. The chemical analysis of planetary atmospheres has become a lively field of modern astrophysics. As you might guess, cosmochemistry depends on spectroscopy—the analysis of light by means of a spectrometer. By exploiting the tools and tactics of spectroscopists, cosmochemists can infer the presence of life on an exoplanet, regardless of whether that life has sentience, intelligence, or technology.

The method works because every element, every molecule—no matter where it exists in the universe—absorbs, emits, reflects, and scatters light in a unique way. And as already discussed, pass that light through a spectrometer, and you'll find features that can rightly be called chemical fingerprints. The most visible fingerprints are made by the chemicals most excited by the pressure and temperature of their environment. Planetary atmospheres are rich with such features. And if a planet is teeming with flora and fauna, its atmosphere will be rich with biomarkers—spectral evidence of life. Whether biogenic (produced by any or all life-forms), anthropogenic (produced by the widespread species *Homo sapiens*), or technogenic (produced only by technology), such rampant evidence will be hard to conceal.

Unless they happen to be born with built-in spectroscopic sensors, our space-snooping aliens would need to build a spectrometer to read our fingerprints. But above all, Earth would have to cross in front of the Sun (or some other source), permitting light to pass through our atmosphere and continue on to the aliens. That way, the chemicals in Earth's atmosphere could interact with the light, leaving their marks for all to see.

Some molecules—ammonia, carbon dioxide, water—show up abundantly in the universe, whether life is present or not. But other molecules thrive in the presence of life itself. Another readily detected biomarker is Earth's sustained level of the molecule methane, two-thirds of which is produced by human-related activities such as fuel oil production, rice cultivation, sewage, and the burps and farts of domestic livestock. Natural sources, comprising the remaining third, include